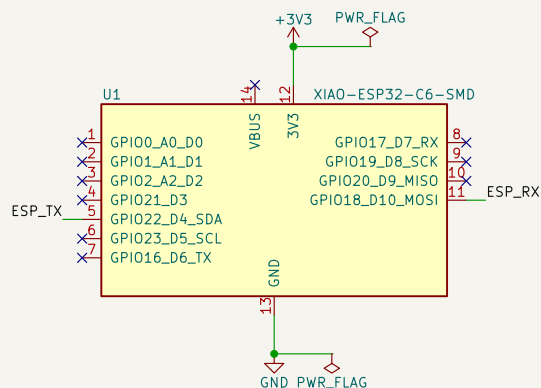


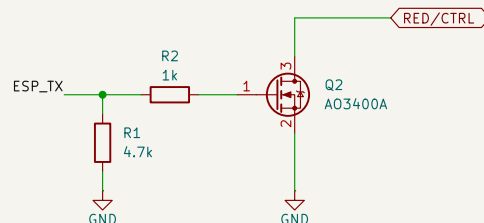
MCU – SEEED STUDIO XIAO ESP32-C6



Powered from the XIAO's own USB-C port (no board power input).
ESP_TX = D4 / GPIO22 (uart_tx_pin)
ESP_RX = D10 / GPIO18 (uart_rx_pin)

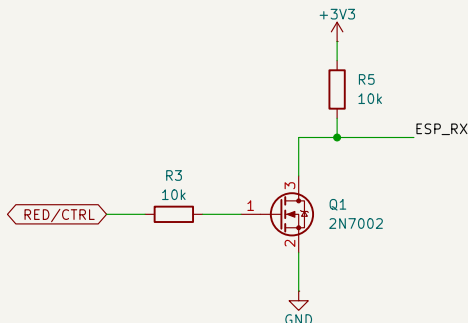
Underside pads 24–33 (MTDI, MTDO, EN, GND, MTMS, MTCK, BOOT, 3V3, BAT, GND) are not soldered and are omitted from this symbol. The PCB footprint marks them with no-via keep-outs.

TX – OPEN-DRAIN LINE DRIVER



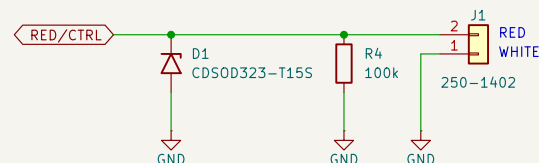
GPIO22 high turns Q2 on and pulls RED low (inverted UART TX).
R1 holds Q2 off while the ESP32 is in reset or unpowered.

RX – INVERTING LEVEL SHIFTER



RED high turns Q1 on and pulls ESP_RX low (inverted UART RX).
R5 is required: gdolib does not enable the internal pull-up.

GDO INTERFACE – SECURITY+ 2.0



J1 lands on the opener's wall-control terminals:
RED – Security+ 2.0 serial data (about 12.7 V idle)
WHITE – ground
D1 clamps transients on the line (15 V standoff).
R4 is a weak bias that defines the line when idle.
Obstruction status comes from the serial protocol;
the black obstruction wire is not used.

NOTES

1. Firmware: ESPHome secplus_gdo (gdolib) with invert_uart = true; uart_tx_pin = GPIO22, uart_rx_pin = GPIO18.
2. Both FET stages invert the signal; the firmware's UART inversion restores the logic sense.
3. Q1's gate sees the RED line directly; use a part rated +/-20 V Vgs (do not substitute AO3400A, +/-12 V).
4. Ordering data: Manufacturer / MPN / Digi-Key PN symbol fields and c6gdo-bom.csv.

ESPHome secplus_gdo: TX GPIO22 (D4), RX GPIO18 (D10), inverted UART
Derived from the rat-ratgdo D1 Mini – ESP32 board

MFLAGG

Sheet: /
File: c6gdo.kicad_sch

Title: XIAO ESP32-C6 Security+ 2.0 Garage Door Interface

Size: A4	Date: 2026-09-16	Rev: 1.0
KiCad E.D.A. 10.0.6		Id: 1/1